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/*main()
{
    //• Growable lists and fixed-length lists
    List<int> id =[101,102,103,104];
    print(id)  ;

    //• Adding elements using add()
id.add(50);
print("After add  :$id");

//• Removing elements using remove()

id.remove(103);
print("After remove :$id");

//• Inserting elements using insert()

id.insert(3, 500);
print("After insert :$id");

//• Sorting numeric or string lists

id.sort();
print("After sort :$id");

// 3: Practicing Sets (Unique Elements)

Set<int>  room_no  ={2,3,4,3,4,5,1,2,5,7,8,4,5,98,};

    print(room_no);

    // add element
    room_no.add(100) ;
    print(room_no);
    //remove
    room_no.remove(100);
    print(room_no);

    Set<int>  teacher_id  ={1,2,3,4,5,6,7,8,9,10};

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//union
print(room_no.union(teacher_id));
//intersection
print(room_no.intersection(teacher_id));
//difference
print(room_no.difference( teacher_id));

//Step 4: Practicing Maps (Key-Value Pairs)

Map<String,String>  name_grade={
    "Miel":"A+",
    "arif":"B+",
    "anas":"A+"
};

print(name_grade);

// Working with Nested Maps

Map<int, Map<String,dynamic>> student={
    101:{"name":"Miel","grade":"A+","Adress":"Dhaka"},
    102:{"name":"arif","grade":"B+"},
    103:{"name":"anas","grade":"A+"}
};

print(student);

//Step 6: Using Map methods and collection operators

print(student.keys);
print(student.values);

Map<String,Map<dynamic,dynamic>> Student_project={
    "Miel Mahmud Sifat":{"Roll":201,"Grade":"A+"},

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        "Arif Billah RIfat":{"Roll":501,"Grade":"A"},

        "Ratul "":{"Roll":201,"Grade":"B+"},

};

}*/

/*
///Step 7: Planning the Student Info System mini project

main()
{
    List<Map<String,dynamic>> Student =[];

    void addStudent(String name,int roll,String grade)
    {
        Student.add({"name":name,"roll":roll,"grade":grade});
    }

    void display ()
    {
        for(var students in Student)
        {
            print("Name:${students['name']}, Roll:${students['roll']},
Grade:${students['grade']}");
        }
    }

    void update_student(int roll,String name,String grade)
    {
        for(var students in Student)
        {
            if(students['roll']==roll)

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        {
            students['name']=name;
            students['grade']=grade;

        }
    }

}

void removeStudent(int roll) {
    Student.removeWhere((student) => student["roll"] == roll);
}

addStudent("Miel", 20, "A+");
addStudent("Arif", 240, "B+");
addStudent("Ratul", 201, "A");
addStudent("John", 202, "B");

display();
removeStudent(201);
print("After removing student with roll 201:");
display();
}

*/

/// 8: Implementing the Student Info System
void main() {
    List<Map<String, dynamic>> students = [
        {
            "name": "Sifat",
            "roll": 101,
            "grade": "A"
        },
        {
            "name": "Abhi",
            "roll": 102,

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        "grade": "B"
    }
];

students.add({
    "name": "Alok",
    "roll": 103,
    "grade": "A+"
});

for (int i = 0; i < students.length; i++) {
    print("Student ${i + 1}");
    print("Name   : ${students[i]["name"]}");
    print("Roll   : ${students[i]["roll"]}");
    print("Grade  : ${students[i]["grade"]}");

}
}
```